

Hamilton cycles in Cayley graphs of polylogarithmic degree

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Abstract

We present a proposed proof that every sufficiently large connected Cayley graph on n vertices and of degree at least $C(\log n)^{13}/\log \log n$ contains a Hamilton cycle. The argument allocates the vertices to translates of a labelled bipartite expander. Edge weights compensate for the unequal allocation probabilities, allowing logarithmic overlap without regularizing the template. Signed rounding preserves balance and expansion, and a sparse matching connects the resulting pieces. Inside each piece, Hamilton routers absorb the matching formed by the paths outside it. A cycle-merging argument then completes the construction. The local routing estimate uses a spectral distance bound after vertex deletion.

Research draft. This is a consolidated proposed proof, not an independently verified theorem. All nonstandard arguments used from the preceding drafts are included below. The classical results used without proof are stated at their points of use. The local absorption argument is an adaptation of the supplied manuscript of Mueyesser [6]; neither the local statement nor the Cayley degree bound is asserted in that manuscript. The present text consolidates the exponent-13 argument [10], rather than claiming a new verification of it.

1 Introduction

We consider the following statement.

Theorem 1.1 (Proposed theorem). *There are absolute constants $C, n_0 > 0$ such that every connected Cayley graph on $n \geq n_0$ vertices and of degree at least*

$$C \frac{(\log n)^{13}}{\log \log n} \tag{1.1}$$

contains a Hamilton cycle.

The main difficulty is that large degree need not imply expansion. An expanding subgraph of large degree need not span the graph, and its translates overlap. Assigning each vertex to one translate introduces another difficulty: a bipartite piece must receive equal numbers of vertices on its two sides. Choosing the pieces first and repairing their imbalances afterwards does not give enough control over the endpoints used to join them.

We impose balance during the allocation. We first find a connected bipartite subgraph F with degrees comparable to the ambient degree, normalized spectral gap of order $(\log n)^{-2}$, and a positive proportion of every retained generator label. We then choose translates of F with logarithmic overlap. Weighted edge incidences give a fractional allocation in which every vertex has total mass one and every copy receives equal mass on its two local sides.

The allocation probabilities need not be almost equal. Instead, we compensate for their variation by assigning weights to the edges of each allocated piece. This makes all weighted degrees have the same initial mean. It also makes the matrix used to control the spectrum have

exactly the required mean. Thus the template need not be regularized, and logarithmic overlap is sufficient. All positive weights stay between two fixed constants. They are only used to analyze the construction; the final cycle uses edges of the original unweighted graph.

The local bipartitions of different copies may disagree. The allocation is therefore rounded in a signed incidence system rather than an ordinary directed network. A cut condition places the fractional allocation in the convex hull of integral allocations. Swap rounding within this convex hull changes at most four coordinates of any fixed copy in a step. A trace-exponential estimate gives the scalar and matrix concentration needed to preserve the weighted degrees and spectral gap. If a retained-label component has odd order, an auxiliary allocation node reserves exactly one vertex in that component. That vertex is later inserted on a path of length two.

We connect the pieces by a locally sparse matching. A useful feature of the construction is that the connecting graph only needs to see the allocated parts directly; it does not need to preserve the spectrum of every full template. We sample generator labels for this purpose, and then select a physical matching by mutual nominations. The matching indicators are negatively associated, so cut concentration has no dependency-graph loss. Marked edges force connectedness, while a signed circulation enforces the endpoint balance.

The final local task is weaker than prescribed spanning paths. Given the matching formed by the paths outside a piece, we only need a Hamilton cycle through that matching. Hamilton routers provide this flexibility. After obtaining a spanning union of cycles, we repeatedly choose a piece meeting more than one cycle and replace its internal path system so that those cycles merge. This avoids prescribing every local pairing in advance.

The proof is organized as follows. Section 2 records the weighted spectral and matching facts. Section 3 proves the local absorption statement. Section 4 gives the signed rounding tools. The weighted partition and the sparse connecting system are constructed in Sections 5 and 6. Section 7 checks the parameters and completes the proposed proof. No theorem from a preceding candidate draft is used as a black box.

2 Preliminaries

We use natural logarithms. Constants c, C may change from line to line. A statement involving sufficiently large order permits enlarging the absolute constant in (1.1). In the Cayley application,

$$X = \text{Cay}(G, S), \quad n = |G|, \quad d = |S|, \quad L = \log n,$$

where $S = S^{-1} \subseteq G \setminus \{1\}$ and $\langle S \rangle = G$. Edges have the form $x \sim xs$, so left multiplication preserves labels. Graphs are simple before contraction. A contracted edge retains its physical identity; parallel edges and loops are allowed, and a loop has two incidences.

For a weighted graph H , write $a_{uv} = a_{vu} > 0$ for the weight of a support edge, $d_H(v) = \sum_u a_{uv}$, and $\text{vol}_H(U) = \sum_{v \in U} d_H(v)$. We write $e_H(U, W)$ for the total weight between disjoint sets. Degrees and edge counts in weighted graphs refer to weights unless called support degrees or physical edge counts. In local statements the positive weights lie in $[\omega, 1]$, for a fixed $\omega > 0$. Constants in those statements may depend on ω .

Let A_H be weighted adjacency, let D_H be the diagonal degree matrix, and put

$$N_H = D_H^{-1/2} A_H D_H^{-1/2}.$$

Adjacency eigenvalues are ordered decreasingly. A normalized gap is always the upper gap $1 - \lambda_2(N_H)$; no lower-eigenvalue bound is imposed. In particular, the eigenvalue -1 in a bipartite graph is allowed.

2.1 Spectral estimates

The variational characterization of the gap is

$$1 - \lambda_2(N_H) = \min_{f \text{ nonconstant}} \frac{\sum_{uv \in E(H)} a_{uv} (f_u - f_v)^2}{\min_z \sum_v d_H(v) (f_v - z)^2}. \quad (2.1)$$

Consequently, a gap of at least σ implies

$$e_H(U, U^c) \geq \sigma \frac{\text{vol}_H(U) \text{vol}_H(U^c)}{\text{vol}_H(V(H))}. \quad (2.2)$$

Both formulas follow by subtracting the degree-weighted mean. We also use the normalized Cheeger inequality: if

$$\Phi(H) = \min_{0 < \text{vol}_H(U) \leq \text{vol}_H(V(H))/2} \frac{e_H(U, U^c)}{\text{vol}_H(U)},$$

then $1 - \lambda_2(N_H) \geq \Phi(H)^2/2$; see [1]. Thus comparable degrees and edge expansion cr/L give normalized gap $c'L^{-2}$. Multiplying every edge weight by factors in a fixed interval changes the gap by at most a fixed factor, directly by (2.1).

Let $u_v = \sqrt{d_H(v)/\text{vol}_H(V(H))}$ and write

$$N_H = uu^\top + E.$$

For the positive part E_+ we have

$$E \preceq E_+, \quad \|E_+\| \leq \max\{0, \lambda_2(N_H)\}, \quad (E_+)^2_{vv} \leq (E^2)_{vv} \leq (N_H^2)_{vv}. \quad (2.3)$$

Indeed, Cauchy–Schwarz gives $(E_+)^2_{vv} \leq (E^2)_{vv}$, spectral calculus gives $E_+^2 \preceq E^2$, and $E^2 = N_H^2 - uu^\top$. If support degrees are between cr and Cr , and all weights are comparable, then every entry of N_H is $O(1/r)$. Hence

$$(E_+)^2_{vv} \leq Cr^{-1/2}. \quad (2.4)$$

This estimate will be applied with $r = d$, even when all the full-copy weights have scale $1/\log n$.

We will repeatedly use the following elementary perturbation observation.

Lemma 2.1. *Suppose a weighted graph has degrees in $[cD, CD]$ and normalized upper gap σ . Delete vertices or edges so that every remaining vertex loses at most $b \leq c'D$ weighted degree. The remaining graph has normalized upper gap at least $\sigma - O(b/D)$.*

Proof. For vertex deletion, first restrict the normalized adjacency matrix to the remaining coordinates. Its second eigenvalue is at most the original second eigenvalue, by interlacing. For edge deletion, the deleted adjacency matrix in the old normalization has norm $O(b/D)$: its absolute row sums are $O(b/D)$. Finally, replacing the old reference degrees by the remaining degrees conjugates the matrix by a diagonal matrix $I + O(b/D)$. The reference-normalized matrix has bounded norm, so this changes it by $O(b/D)$. $\square$

2.2 Sampling matrices

For independent centered self-adjoint matrices Y_j of dimension r , we use matrix Bernstein [7] in the form

$$\mathbb{P}\left(\left\|\sum_j Y_j\right\| \geq t\right) \leq 2r \exp\left(-\frac{t^2}{2v + 2bt/3}\right), \quad \|Y_j\| \leq b, \quad \left\|\sum_j \mathbb{E}Y_j^2\right\| \leq v. \quad (2.5)$$

We also use scalar Chernoff bounds, including their without-replacement versions for uniform subsets. For nonnegative weights bounded by a constant, their exponents have the usual order $t^2/(\mu + t)$, where μ is the mean.

The next column-sampling estimate, with its proof, is the one used in the supplied paper [6, Lemma 2.4].

Lemma 2.2. *Let F be an $r \times s$ real matrix, let $c = \max_j \|Fe_j\|$, and let I be a uniform k -subset of its columns. Put $q = k/s$. For $t > 0$,*

$$\mathbb{P}(\|F[:, I]\| > \sqrt{q} \|F\| + c\sqrt{t}) \leq r(s+1)e^{-t}.$$

Proof. First retain columns independently with probability q . If the columns are f_j and their indicators are ξ_j , then

$$F[:, I]F[:, I]^\top = \sum_j \xi_j f_j f_j^\top.$$

The centered summands have norm at most c^2 , and their variance sum is at most $qc^2FF^\top$ in the semidefinite order. The one-sided version of (2.5) gives, except with probability re^{-t} ,

$$\|F[:, I]\|^2 \leq q\|F\|^2 + \sqrt{2q}c\|F\|\sqrt{t} + \frac{2}{3}c^2t \leq (\sqrt{q}\|F\| + c\sqrt{t})^2.$$

For $q = k/s$, the value k is a mode of $\text{Bin}(s, q)$, so its probability is at least $1/(s+1)$. Conditioning on exactly k retained columns proves the assertion. The cases $k = 0$, s or $c = 0$ are immediate. $\square$

Lemma 2.3 (Bipartite sampling). *Let H have bipartition classes of size N , weights in $[\omega, 1]$, degrees $(1 \pm \eta)D$, and normalized upper gap σ , where $\eta \leq c\sigma$. Let $L \geq \log(2N)$. Choose independent uniform sets of size pN from the two classes. If*

$$pD \geq C\sigma^{-2}L, \tag{2.6}$$

then, with failure probability at most e^{-aL} , the graph between the sets has degrees $(1 \pm c_1\sigma)pD$ and normalized upper gap at least $c_2\sigma$. Here a can be any fixed constant by increasing C , and c_1 can be made sufficiently small compared with c_2 .

Proof. Let K be the biadjacency matrix and let d_A, d_B be its degree vectors. Put

$$R_0 = \frac{d_A d_B^\top}{e(H)}, \quad F = K - R_0.$$

The normalized biadjacency matrix has top singular value one and all other singular values at most $1 - \sigma$. Undoing its degree normalization gives

$$\|F\| \leq (1 + \eta)(1 - \sigma)D, \quad \max\{\text{row and column squared norms of } F\} \leq CD. \tag{2.7}$$

For the second bound, use $\sum_j K_{ij}^2 \leq d_H(i)$ and $D \leq N$ to bound the rank-one term as well.

Apply Lemma 2.2 to sample the rows. Simultaneously, scalar concentration gives

$$\|F[U, B]\| \leq \sqrt{p}\|F\| + C\sqrt{DL}, \quad \|F[U, \{v\}]\|^2 \leq CpD \quad (v \in B).$$

The second inequality follows from $\sum_{u \in U} K_{uv}^2 \leq d_H(v, U)$ and the explicit rank-one term. Conditional on these events, another column-sampling application yields

$$\|F[U, V]\| \leq p\|F\| + C\sqrt{pDL}. \tag{2.8}$$

The rank-one term does not affect the second singular value. By (2.6), the error is a sufficiently small multiple of σpD . Scalar concentration gives the asserted degree estimates. Replacing reference degree pD by actual degrees perturbs the normalized matrix by a sufficiently small multiple of σ . This proves the gap assertion. Choosing the tail constants first gives the stated failure probability. $\square$

We record explicitly the conditional form needed when some vertices have already been used.

Lemma 2.4 (Fixed boundary, fresh layer). *In the setting of Lemma 2.3, suppose a fixed set U of pN rows satisfies*

$$\|F[U, B]\| \leq \sqrt{p} \|F\| + C\sqrt{DL}, \quad \|F[U, \{v\}]\|^2 \leq CpD, \quad d_H(v, U) = (1 \pm c_1\sigma)pD \quad (2.9)$$

for every opposite-class vertex v . Let B' be any remaining column pool with $|B'| = (1 - O(q))N$, where $q \leq c\sigma$, and suppose every vertex has lost at most $c\sigma D$ weighted neighbours from that pool. A fresh uniform pN -subset V of B' has the degree and gap conclusions of Lemma 2.3, with suitable fixed constants.

Proof. Deleting columns cannot increase the norm of $F[U, B]$. Put $p' = pN/|B'| = p(1 + O(q))$. Lemma 2.2, conditional on the entire choice of B' , gives

$$\|F[U, V]\| \leq \sqrt{pp'} \|F\| + C\sqrt{pDL}.$$

The factor $1 + O(q)$ fits inside the original gap. The degrees from V to U are controlled by (2.9), and those from U to V concentrate about $p'd_H(u, B') = (1 \pm c_3\sigma)pD$. The preceding normalization argument applies. The conclusion is uniform over all pools satisfying the displayed deterministic conditions; no independence of the earlier deletion is asserted. $\square$

2.3 Perfect matchings with a small enlargement

Call an h -vertex weighted graph ξ -cut-dense if

$$e_H(T, T^c) \geq \xi|T||T^c| \quad (T \subseteq V(H)).$$

Equation (2.2) shows that comparable degrees D and gap σ imply cut-density $c\sigma D/h$.

We will use the following weighted form of the counting argument in [6, Lemma 5.1].

Lemma 2.5 (Robust Hall). *Let H_0 be a balanced bipartite weighted graph with classes L_0, R_0 of size m , degrees $(1 \pm \eta)D_0$, and cut-density ξ . Put $\alpha = \xi m/D_0$, and suppose $0 < \alpha \leq 1/2$ and $\eta \leq \alpha/10$. Add sets X, Y of equal size to the two respective classes. Suppose each added vertex has at least $D_0/2$ weighted neighbours in the opposite old class, while each old vertex has at most $\alpha D_0/20$ weighted neighbours in the opposite added class. Then the support of the enlarged graph has a perfect matching.*

Proof. For $S \subseteq L_0, T \subseteq R_0$ with no edge between them, set $s = |S|, t = |T|$ and $h = m - s - t$. Assume $s \leq t$. The set $Z = S \cup (R_0 \setminus T)$ has size $2s + h \leq m$. Cut-density and the degree bounds give

$$\alpha D_0(2s + h) \leq e_{H_0}(Z, Z^c) \leq D_0(2\eta s + (1 + \eta)h).$$

It follows that

$$m - |S| - |T| \geq \alpha \min\{|S|, |T|\}. \quad (2.10)$$

In particular, with no added vertices, Hall's theorem gives a perfect matching.

If the enlarged graph has no perfect matching, Hall's theorem supplies edge-free sets $S' \subseteq L_0 \cup X$ and $T' \subseteq R_0 \cup Y$ with $|S'| + |T'| > m + |X|$. Write $S = S' \cap L_0, T = T' \cap R_0, x = |S' \cap X|, y = |T' \cap Y|$, and $h = m - |S| - |T|$. Then $h < \min\{x, y\}$. Counting the edges from $S' \cap X$ into $R_0 \setminus T$, and similarly on the other side, gives

$$x \leq \frac{\alpha}{10}(m - |T|), \quad y \leq \frac{\alpha}{10}(m - |S|).$$

With $u = \min\{|S|, |T|\}$, these inequalities and (2.10) yield

$$h < \frac{\alpha}{10}(h + u) \leq \frac{\alpha+1}{10}h,$$

a contradiction. $\square$

3 Local absorption

The local object must accommodate the matching formed by the paths outside a piece. That matching need not respect the local bipartition.

Definition 3.1. Let $E \subseteq V(H)$ have positive even size. We say that (H, E) is *matching-absorbing* if, for every perfect matching J on E , the multigraph $H + J$ has a Hamilton cycle containing every edge of J . Edges of J are formal distinguished edges, kept separate from any parallel edge of H .

Here is the local statement used later.

Theorem 3.2 (Local absorption). *Fix $\omega > 0$. There are constants $c, C > 0$ with the following property. Let H be bipartite with classes of size N , weights in $[\omega, 1]$, degrees $(1 \pm \eta)D$, and normalized upper gap at least $\sigma \in (0, 1/10]$, where $\eta \leq c\sigma$. Let $L \geq \max\{\log(2N), 10\}$, and let $g \geq 4$ satisfy*

$$D \geq C\sigma^{-9/2}L^2g, \quad g \geq \frac{CL}{\log(2 + \sigma^{5/2}D/L)}. \quad (3.1)$$

If E contains $\ell \geq 1$ vertices of each class and

$$\ell \leq c \frac{\sigma^{5/2}N}{Lg}, \quad \max_v d_H(v, E) \leq c \frac{\sigma^{5/2}D}{L}, \quad (3.2)$$

then $(\text{supp } H, E)$ is matching-absorbing.

We prove the theorem in several steps. The first is a short-path estimate that uses the spectral gap directly, rather than passing through a vertex-expansion bound.

3.1 Short connections after deletion

Lemma 3.3 (Distances after deletion). *Let F be a weighted graph on M vertices, with degrees between aD and bD for fixed $a, b > 0$, and normalized upper gap at least $\sigma \in (0, 1]$. There are constants $c, C > 0$ such that, whenever $|U| \leq c\sigma M$, there is a set $K \subseteq V(F) \setminus U$ with*

$$|V(F) \setminus K| \leq C|U|/\sigma$$

such that every two vertices of K are joined in $F - U$ by a path of length at most $C\sigma^{-1/2}\log(2M)$. The paths need not remain in K .

Proof. Let u be the stationary unit vector of N_F and put $P = (I + N_F)/2$. Then $P \succeq 0$, $Pu = u$, and $\lambda_2(P) \leq 1 - \sigma/2$. Let Q project onto U^c and write $\tau = \|u_U\|^2 \leq C|U|/M$. On the remaining coordinates, the largest eigenvalue θ of $P_U = QPQ$ satisfies

$$\theta \geq \frac{u_{U^c}^\top P u_{U^c}}{1 - \tau} \geq \frac{1 - 2\tau}{1 - \tau} \geq 1 - 2\tau.$$

Here nonnegativity of P gives the middle inequality, and $\tau \leq 1/2$ gives the last.

Choose a nonnegative unit top eigenvector f , extended by zero on U , and write $f = \alpha u + z$ with $z \perp u$. Then

$$\theta = f^\top P f \leq 1 - \frac{\sigma}{2}(1 - \alpha^2), \quad \|f - u\|^2 \leq 2(1 - \alpha^2) \leq C\tau/\sigma.$$

Every coordinate of u is at least $cM^{-1/2}$. Thus all but $C|U|/\sigma$ vertices satisfy $f_v \geq u_v/2$. Let K be these vertices.

For c sufficiently small, $\theta \geq 1 - \sigma/4$. Positivity and interlacing place every other eigenvalue of P_U in $[0, b_0]$, where $b_0 = 1 - \sigma/2$. With T_r denoting the Chebyshev polynomial, define

$$q_r(t) = \frac{T_r(2t/b_0 - 1)}{T_r(2\theta/b_0 - 1)}.$$

The identities $|T_r(x)| \leq 1$ for $x \in [-1, 1]$ and $T_r(\cosh s) = \cosh(rs)$ show that $q_r(\theta) = 1$ and

$$\max_{0 \leq t \leq b_0} |q_r(t)| \leq 2e^{-cr\sqrt{\sigma}}.$$

Therefore $\|q_r(P_U) - ff^\top\| \leq 2e^{-cr\sqrt{\sigma}}$. For $r = \lceil C\sigma^{-1/2} \log(2M) \rceil$, this is smaller than half the least value of $f_x f_y$ for $x, y \in K$. Hence $q_r(P_U)_{xy} > 0$ for these vertices. An off-diagonal entry of a polynomial of degree r vanishes when the support distance exceeds r . The desired paths follow. The use of P rather than N_F is what permits a bipartite host. $\square$

We use Haxell's matching theorem in the following form [2]. If a hypergraph has vertex partition $A \cup B$, every edge meets A once and B at most $r - 1$ times, and the family of B -parts of edges meeting $I \subseteq A$ has vertex-cover number greater than $(2r - 3)(|I| - 1)$ for every nonempty I , then there is a matching saturating A .

Lemma 3.4 (Spectral connection). *Suppose the weighted graph induced by a reservoir R has comparable degrees with maximum Δ and normalized upper gap at least σ . Let $L \geq \log(2|R|)$, and let (a_j, b_j) be pairs of distinct vertices outside R , each vertex occurring in at most ten pairs. Let T be their endpoint set. Suppose*

$$d(x, R) \geq \beta\Delta \quad (x \in T), \quad d(v, T) \leq c\beta\sigma^{3/2}\Delta/L \quad (v \in R), \quad (3.3)$$

where $0 < \beta < 1$. Then the pairs can be joined by paths with disjoint internal vertex sets in R , each of length at most $CL/\sqrt{\sigma}$.

Proof. Put $s = \lceil C_0 L/\sqrt{\sigma} \rceil$, taking C_0 sufficiently large to allow the two endpoint edges. For each pair, form the hyperedges consisting of its index and the internal vertices of all its paths of length at most s through R . By Haxell's theorem, it suffices to show that, for every nonempty set I of indices and every $U \subseteq R$ with $|U| \leq 2s(|I| - 1)$, one indexed pair has such a path avoiding U .

Write $\tau = c\beta\sigma^{3/2}\Delta/L$. Counting weighted incidences gives

$$2\beta\Delta|I| \leq 10|R|\tau, \quad |I| \leq 5c\sigma^{3/2}|R|/L.$$

Thus $|U| \leq Cc\sigma|R|$, and Lemma 3.3 applies. It supplies K with $|R \setminus K| \leq C|U|/\sigma$, whose vertices are mutually reachable by short paths in $R - U$.

If no pair has both endpoints adjacent to K , choose from every indexed pair an endpoint whose reservoir neighbourhood lies in $R \setminus K$. The multiplicity bound then gives

$$\beta\Delta|I| \leq 10\tau|R \setminus K| \leq C\tau|U|/\sigma \leq C\tau s|I|/\sigma < \beta\Delta|I|,$$

provided c is sufficiently small. For $U = \emptyset$ the same argument has $K = R$. Hence some pair has neighbours in K , and these neighbours can be joined in $R - U$ within the allowed length. Haxell's theorem finishes the proof. $\square$

3.2 Comparators and Hamilton routers

A graph S with disjoint sets I, O of size w is an I, O -Hamilton router if, for every bijection from O to I , it has a spanning partition into w paths between I and O whose union with the corresponding formal matching is a Hamilton cycle. A *comparator* has two input and two output vertices and admits spanning path partitions realizing both input-output bijections. These are the router notions used in [6].

Lemma 3.5 (Even-cycle comparator). *Every even cycle can be completed to a comparator by choosing four terminals and pairing the remaining cycle vertices, then adding internally disjoint paths for those pairs. Each added path meets the cycle only at its two endpoints. Its length may be arbitrary, provided it is realized in the host graph.*

Proof. A 4-cycle is already a comparator, with opposite vertices as inputs. For a cycle $c_0c_1 \cdots c_{2r-1}c_0$, $r \geq 3$, take inputs c_0, c_2 and outputs c_1, c_4 . Put

$$M_0 = \{c_{2j}c_{2j+1} : 0 \leq j < r\}, \quad M_1 = \{c_{2j+1}c_{2j+2} : 0 \leq j < r\},$$

where the indices in M_1 are cyclic, and pair the remaining vertices by

$$P = \{c_{2j+1}c_{2j+4} : 1 \leq j \leq r-3\} \cup \{c_{2r-3}c_{2r-1}\}. \quad (3.4)$$

This is the construction from [6, Lemma 4.5], with the cycle matchings named consistently.

The graph $M_0 \cup P$ consists of the edge c_0c_1 and a path from c_2 to c_4 through all remaining vertices. Indeed, contract the other M_0 -edges and label the resulting vertices $1, \dots, r-1$. The edges of P are $j(j+2)$ for $1 \leq j \leq r-3$, together with $(r-2)(r-1)$. They form a path with endpoints 1, 2. Similarly, $M_1 \cup P$ consists of c_1c_2 and a path from c_0 to c_4 : after contracting the other M_1 -edges, the pairs in (3.4) form the path $1, 2, \dots, r-1$. Replacing the edges of P by the prescribed disjoint paths preserves both spanning path partitions. $\square$

Lemma 3.6 (Linear-size router). *For every $w \geq 2$ there is a Hamilton-router template with $w-1$ comparators and $O(w)$ fixed connections. Replacing its comparators by arbitrary comparators and its fixed connections by internally disjoint paths preserves the router property.*

Proof. Start with w directed logical wires from I to O . In succession, place one comparator between wires j and $j+1$, for $1 \leq j < w$. A wire continues from an output terminal to its next input terminal, or to its final output. The total number of connections is $O(w)$. For any choice of comparator settings, the network is partitioned into w spanning I - O paths.

Fix an outside matching from O to I . An initial setting gives a union of cycles. If more than one cycle remains, some comparator has its two chosen paths in different cycles. Otherwise every comparator lies entirely in one cycle, and every fixed connection joins vertices of that same cycle, contradicting connectedness of the whole template. Switching the chosen comparator joins its two cycles into one. Repeating yields a single cycle. The argument depends only on the two path partitions in each comparator and on the fixed connections, so it survives their replacements. $\square$

We need many short cycles, rather than a girth bound for the entire host.

Lemma 3.7. *A bipartite weighted graph on M vertices, with weights in $[\omega, 1]$ and degrees comparable to $D_C \geq C$, contains at least cM/g vertex-disjoint cycles of length at most g , provided*

$$g \geq C \log(2M) / \log(2 + D_C).$$

Proof. Take a maximal packing. If its vertices cover fewer than cM vertices, for a sufficiently small fixed c , deleting them leaves average support degree at least $c'D_C$. Repeated deletion of vertices of degree below half this average leaves a nonempty graph of minimum degree at least $c''D_C$. Breadth-first search in a graph of minimum degree δ and girth greater than $2r+1$ exposes at least $1 + \delta \sum_{j < r} (\delta-1)^j$ vertices. Thus the remaining graph has a cycle of length at most g , a contradiction. The packing therefore covers at least cM vertices and has at least cM/g members. Bipartiteness makes all these cycles even. $\square$

We also use that a graph of comparable degrees and normalized gap σ has a path with at least $c\sigma M$ vertices. To see this, run depth-first search. When the completed set first has $\lfloor M/3 \rfloor$ vertices, it has no edge to the unexplored set. Its boundary is in the current stack, which is a path. Equation (2.2) and the maximum-degree bound force that stack to have $c\sigma M$ vertices, unless it was already longer than $M/6$.

3.3 Embedding the router and attaching the endpoints

Proof of Theorem 3.2. Fix a perfect matching J on E . We construct a Hamilton cycle containing it. The random choices may depend on J ; no union bound over all matchings is needed.

Delete E for the initial random partition and call the balanced remaining graph H_0 . Put $N_0 = N - \ell$. By (3.2) and Lemma 2.1, its degrees remain $(1 \pm c'\sigma)D$ and its gap is at least a fixed positive multiple of σ . We absorb these fixed changes by decreasing the small constants below.

Choose small constants in the order

$$0 < c_1 \ll c_3 \ll c_0 \ll 1,$$

and put $q = c_0\sigma$. Let w be the largest even integer at most

$$c_1 \frac{\sigma^{5/2} N_0}{Lg}, \quad m = w/2. \quad (3.5)$$

The constant in (3.2) is chosen so that $\ell \leq w/8$. The degree assumption and $D \leq N$ give $w \geq cC\sigma^{-2}L$, so rounding these quantities loses only fixed factors.

Partition each original colour class uniformly into sets with the following prescribed sizes, using independent partitions in the two classes. Reserve three balanced sets R_1, R_2, Z , each containing $\lfloor qN_0 \rfloor$ vertices of each colour. Choose disjoint mixed-colour sets I, O, U, V , each containing m vertices of each colour. The remaining vertices are left in a pool. Here I, O are router terminals, R_1 is for endpoint attachments, R_2 for the router, and Z is for divisibility. All set sizes are integral from now on.

Orient the edges of J arbitrarily. If the tails contain a positive and b negative endpoints, then the heads contain b positive and a negative endpoints, and $a + b = \ell$. Within U , choose uniformly a positive and b negative new ports for the tails. Within V , choose b positive and a negative new ports for the heads. Assign the ports to the corresponding original endpoints by uniform bijections within each colour and role. These choices can equivalently be included as cells of the initial two colour partitions. In particular, the union P of the new ports is a uniform ℓ -set of each colour.

Within R_2 , choose a balanced set C with

$$|C \cap A| = |C \cap B| = \left\lfloor c_3 \sigma^{3/2} q N_0 / L \right\rfloor, \quad (3.6)$$

and let $Q = R_2 \setminus C$. We first specify events that hold simultaneously with positive probability, before any paths are chosen.

The graphs on R_1, Q, Z have comparable weighted degrees of order qD and gap $c\sigma$, by Lemma 2.3. Every original vertex has its expected weighted degree, within a small fixed relative error, into each of these sets. The graph on C has degrees comparable to

$$D_C = \Theta(\sigma^{3/2} q D / L) = \Theta(\sigma^{5/2} D / L), \quad (3.7)$$

and every vertex has at most CD_C neighbours by weight in C . The lower bound in (3.1) gives $D_C \geq cC\sigma^{-2}Lg$, so all the degree estimates have exponential tails sufficient for a union bound.

For each colour block of I, O, U, V , impose the one-sided events (2.9), including degree concentration for every vertex of H . Impose them as well for the finitely many filler blocks in the special transition $U-V$, as described below. Their sizes are at least $w/4$ and

$$\frac{wD}{N_0} = \Theta\left(\frac{\sigma^{5/2} D}{Lg}\right) \geq cC\sigma^{-2}L. \quad (3.8)$$

Thus Lemmas 2.2 and 2.3 apply with sufficiently small degree error compared with the retained gap.

Finally impose the degree bounds

$$d_H(v, P) \leq C\ell D/N_0 + CL, \quad (3.9)$$

$$d_H(v, \mathcal{B}) \leq CqD + d_H(v, E) \leq c_4\sigma D, \quad \mathcal{B} = E \cup R_1 \cup R_2 \cup Z \cup I \cup O \cup U \cup V. \quad (3.10)$$

The constant c_4 can be made as small as needed by choosing c_0 small. The set $\mathcal{B}$ is fixed before the adaptive choices below; it contains every vertex that can be removed from the eventual fresh-layer pool. All original vertices, including those in E , are included in the scalar tests. Ordinary concentration and a union bound establish all the events just listed. Fix a partition satisfying them.

By Lemma 3.7, the graph on C contains at least $c|C|/g$ disjoint cycles of length at most g . The second condition in (3.1), together with (3.7), guarantees the required cycle-length bound. Since $c_1 \ll c_3$, there are at least $w - 1$ cycles. Turn them into the comparators from Lemma 3.5, and arrange them according to Lemma 3.6.

All auxiliary connection endpoints lie in $C \cup I \cup O$, and each is used only a bounded number of times. Every endpoint has cqD weighted neighbours in Q . On the other hand, every vertex of Q sees at most

$$CD_C + CwD/N_0 + CL \leq c_5\sigma^{3/2}qD/L$$

weighted neighbours in the whole endpoint set. Choosing c_3, c_1 sufficiently small makes Lemma 3.4 applicable. It supplies all the internally disjoint paths required for the comparators and their fixed connections. Their union is an I, O -Hamilton router $S \subseteq R_2 \cup I \cup O$.

Although its terminal sets are mixed-colour, S is balanced in the original bipartition. Every comparator setting gives a spanning path partition of S , and the difference between its two colour counts equals half the sum of the signs of its endpoints. The latter is zero because $I \cup O$ is balanced.

Next join each original endpoint in E to its designated new port of the same colour, using R_1 for the internal vertices. The original endpoint bound and (3.9) imply

$$\max_{v \in R_1} d_H(v, E \cup P) \leq c_6\sigma^{5/2}D/L = c'_6\sigma^{3/2}qD/L.$$

Every endpoint has cqD neighbours in R_1 . Thus Lemma 3.4 provides all the attachments with disjoint interiors. An attachment between two vertices of the same colour has even length. Deleting all its vertices except its new port therefore removes equal numbers of the two colours.

Replace each edge of J , together with its two attachments, by a formal edge between the corresponding new ports. Denote the resulting formal matching between U and V by J' . The removed vertices are balanced, and the actual attachment paths are retained for the final expansion. None meets the router. We have only used paths inside the previously fixed envelope (3.10).

Divisibility and the remaining layers

Let N' be the number of vertices of each colour outside S after the original endpoints and attachment interiors have been removed. Both U and V are still present. Write r for the residue of N' modulo m , so $0 \leq r < m$.

If $r > 0$, use the graph on Z to find a path of length $2r + 1$. The depth-first-search observation preceding this proof gives a path with $c\sigma|Z|$ vertices, and

$$c\sigma|Z| \gg w > 2r.$$

Its interior has r vertices of each colour. Replace it temporarily by a formal edge between its positive and negative endpoints, placing those endpoints in $U \cap A$ and $V \cap B$, respectively. Evict one predesignated non-port vertex from each of those blocks into the remaining pool. The two dummy vertices are chosen as part of the initial random partition; they are never original endpoints or new ports. If $r = 0$, make no change.

The number of vertices of each colour outside S , after this removal of path interiors, is $N'' = N' - r$, divisible by m . The special layers U, V still have m vertices of each colour. Partition the remaining balanced pool uniformly into mixed layers $F_1, \dots, F_t$, each with m vertices of each colour. There are at least two such layers, since $w/N_0 = o(1)$. Order the layers as

$$O, F_1, U, V, F_2, \dots, F_t, I.$$

We find a perfect matching between each consecutive pair of layers. Each ordinary transition separates into the two cross-colour bipartite matching problems.

For transitions between two fresh layers, the residual host has comparable degrees $(1 \pm c_7\sigma)D$ and gap $c_8\sigma$, by (3.10) and Lemma 2.1. Lemma 2.3 applies by (3.8). Transitions involving I, O, U, V use Lemma 2.4 and the initial one-sided events. The size of the fresh pool is $(1 - O(q))N_0$ in each colour, and (3.10) supplies the required degree-loss bound. The constants are chosen so that degree errors are smaller than the cut-density parameter. Lemma 2.5, with no enlargement, then gives perfect matchings.

If a vertex of U or V has been replaced for divisibility, first consider the transition using its old dummy vertex. After establishing the usual sampled graph estimates, delete that dummy and one arbitrary vertex from the opposite fresh block. The resulting balanced core loses at most one neighbour at every vertex, so it retains its degree and cut-density bounds. The new path endpoint and the deleted fresh vertex each have at least half the reference degree into the opposite core: for the former this follows from fresh degree concentration for all vertices, and for the latter from the original one-sided degree bounds. Each core vertex sees at most one added neighbour by weight. Lemma 2.5 adds these two vertices back. This justifies the transitions adjacent to the modified special layer.

It remains to handle $U-V$. Include every edge of J' and the optional divisibility edge. After their endpoints are removed, the two physical cross-colour blocks are balanced. Indeed, the $U \cap A$ tail count and the $V \cap B$ head count are both a , while the other two counts are both b . The optional new edge removes one additional vertex from each of $U \cap A$ and $V \cap B$.

Before any adaptive choice, the initial events required perfect matchings in the two random filler blocks, both with and without the designated dummy vertices. Each block has at least $w/4$ vertices on each side. The two colour partitions are independent, so these are precisely the random bipartite graphs covered by Lemma 2.3 and Lemma 2.5. If divisibility is used, the old dummy vertices have simply been removed from these filler blocks, and the new endpoints are covered by the formal edge. Thus a perfect matching exists in the special transition as well.

There are at most $2N$ transitions. Taking sufficiently strong tails in the sampling lemmas and then a union bound gives a successful fresh equipartition, conditional on all earlier choices. This conditioning is legitimate: the new pool has already been fixed, and the initial events for every fixed boundary are deterministic inequalities valid for that pool. No adaptive leftover is treated as an independent sample of the original host.

The union of the transition matchings is a collection of w disjoint $O-I$ paths whose internal vertices are precisely those outside S after the temporary path removals. The Hamilton-router property joins them through S into one cycle. Expand the divisibility edge and then every edge of J' into its attachment paths and its original edge of J . All the expanded interiors are disjoint. We obtain a Hamilton cycle in $H + J$ containing every edge of J . $\square$

3.4 From local absorption to a single cycle

The following observation is the only global property of the local theorem that will be used.

Lemma 3.8 (Cycle merging). *Suppose $V_1, \dots, V_t$ partition $V(X) \setminus W$. Let $\mathcal{M}$ be a family of edges and paths whose endpoints are mutually distinct and lie in the parts, and whose disjoint interiors partition W . Let E_i be its endpoint set in V_i . Suppose every E_i is nonempty, the contraction of $\mathcal{M}$ on the parts is connected, and (H_i, E_i) is matching-absorbing for a spanning graph H_i on V_i using no edge of $\mathcal{M}$. Then X has a Hamilton cycle.*

Proof. For each i , take any formal perfect matching on E_i , apply absorption, and delete the formal matching. This gives a spanning path system in V_i . Combining the systems with $\mathcal{M}$ gives a spanning union of disjoint cycles.

If more than one cycle remains, some part meets at least two cycles. Otherwise each part belongs to just one cycle and every contracted connecting edge joins parts in that same cycle, contrary to connectedness. Delete the current internal paths of such a part V_i from all cycles meeting it. The remaining outside paths pair E_i , and hence define a formal perfect matching J_i . Apply absorption to J_i and substitute its actual outside paths for the formal edges. All cycles meeting V_i merge into one; other cycles remain unchanged. Iteration gives the required Hamilton cycle. $\square$

4 Signed rounding

A signed multigraph has a sign in $\{+1, -1\}$ at each end of each edge. Its signed incidence matrix B has one row per vertex; the column of an edge is the sum of the two signed unit vectors corresponding to its ends. A loop therefore has column $2e_v$, $-2e_v$, or zero. We write $\delta(U)$ for an ordinary cut of the underlying multigraph, so loops never belong to a cut. The graph is *balanced* if every vertex has equally many positive and negative incidences.

Signed incidence matrices need not be totally unimodular. We first give the particular integrality criterion that will replace directed-flow integrality.

4.1 Integrality from cut slack

We use the classical unit-capacitated perfect b -matching polytope description; see [5, Section 1]. For a loopless multigraph and a nonnegative integral demand vector b , the convex hull of edge sets having degree b_v at every vertex consists of the vectors y satisfying the degree equations, $0 \leq y_e \leq 1$, and

$$y(\delta(U) \setminus J) + (1 - y)(J) \geq 1 \quad \text{if } J \subseteq \delta(U) \text{ and } b(U) + |J| \text{ is odd.} \quad (4.1)$$

Here $(1 - y)(J) = \sum_{e \in J} (1 - y_e)$. This polytope theorem, rather than total unimodularity, is the standard input in the next proof.

Lemma 4.1 (Robust signed integrality). *Let B be a signed incidence matrix, let $b \in \mathbb{Z}^V$, and suppose $\sum_{v \in K} b_v$ is even for every connected component K of the underlying graph. If $Bx = b$, $x \in [0, 1]^E$, and*

$$\sum_{e \in \delta(U)} \min\{x_e, 1 - x_e\} \geq 1 \quad (\emptyset \neq U \neq K) \quad (4.2)$$

within every component, then

$$x \in \text{conv}\{z \in \{0, 1\}^E : Bz = b\}.$$

Proof. Work in one component. Convert the signed system to an ordinary unit-capacitated degree system. A $++$ edge retains variable x_e . A $--$ edge receives variable $1 - x_e$. For a mixed-sign edge, insert an auxiliary vertex of demand one and replace the edge by two edges, carrying x_e at its positive end and $1 - x_e$ at its negative end. Increase each original demand by its number of negative incidences.

A resulting same-sign loop with variable y is replaced by a triangle $vabv$ carrying values $y, 1 - y, y$, with demand one at a, b . A mixed-sign loop is represented by two parallel edges to one auxiliary vertex, with values $x_e, 1 - x_e$ and auxiliary demand one. The degree equations at auxiliary vertices force precisely these relations, so the transformation is an affine bijection on feasible sets and on their integral points. The total demand remains even. Fractional feasibility makes every expanded demand nonnegative.

We verify (4.1) for the expanded point. Let U_0 be the original vertices in a proposed blossom shore U . If U_0 is nonempty and proper, every original edge crossing U_0 has a gadget edge crossing U . Its value is x_e or $1 - x_e$, so its contribution to the left side of (4.1), whether or not it belongs to J , is at least $\min\{x_e, 1 - x_e\}$. Gadgets for distinct original edges are edge-disjoint. The slack-cut condition proves the inequality.

If U_0 is empty, the shore contains only auxiliary vertices. Their gadget interiors are disjoint. A mixed-edge gadget, or one auxiliary vertex of a loop triangle, has demand one and two boundary values $y, 1 - y$. Odd local blossom parity means that zero or two of these boundary edges belong to J , giving contribution exactly one. If both auxiliary vertices of a loop triangle belong to U , their demand sum is two and their boundary values are y, y . Odd local parity requires exactly one boundary edge in J , again giving contribution one. Since the total parity is odd, some gadget has odd local parity; all other contributions are nonnegative. Finally, if U_0 is all original vertices, take the complementary shore. The cut is unchanged and total expanded demand is even, so the parity is unchanged.

Thus every blossom inequality holds. The polytope description and the affine correspondence prove the result. $\square$

4.2 Rounding with small support in each row

Lemma 4.2 (Signed swap rounding). *If $x \in \text{conv}\{z \in \{0, 1\}^E : Bz = b\}$, then x can be rounded to one of these integral points by a finite sequence of mean-preserving two-point line moves. Each move changes at most four edge variables incident with any fixed underlying vertex.*

Proof. Fix a finite convex decomposition of x . Consider two binary solutions a, b in it, with masses $\alpha, \beta > 0$. On their disagreement edges write $h = b - a \in \{-1, 1\}$ and multiply both incidence signs of edge e by h_e . Since $Bh = 0$, the resulting signed graph has equally many positive and negative incidences at every vertex. Pair opposite incidences. Together with the edge pairings, these pairs decompose the disagreement edges into alternating closed trails.

Choose an inclusion-minimal nonempty alternating closed trail. It visits each vertex at most twice: among three visits, two have the same incoming sign, and the intervening segment is a smaller alternating closed trail. Thus it uses at most four incidences at any vertex. If δ agrees with h on its edges and is zero elsewhere, then $B\delta = 0$, and $a + \delta, b - \delta$ are binary feasible solutions.

Replace a by $a + \delta$ with probability $\beta/(\alpha + \beta)$, or replace b by $b - \delta$ with probability $\alpha/(\alpha + \beta)$. The current weighted point moves by $\alpha\delta$ or $-\beta\delta$, with conditional mean zero. In either outcome, the two solutions agree on the trail, so their disagreement set shrinks. Continue until they coincide and combine their masses. Merging all solutions in this way gives the required process. $\square$

The next estimate supplies concentration without independence. Its support restriction is applied to the variables of a single copy, not to the full allocation at once.

Lemma 4.3 (Bounded-support concentration). *Let $x^{(0)}, \dots, x^{(T)}$ be a finite randomized process in $[0, 1]^s$. At each step, conditional on the past, it moves to one of two points on a line through its current point, with unchanged mean. Suppose each direction is supported on at most four coordinates. For positive semidefinite $q \times q$ matrices A_j with $\|A_j\| \leq b$, put*

$$M = \sum_j x_j^{(0)} A_j, \quad Z = \sum_j x_j^{(T)} A_j.$$

If $\|M\| \leq \nu$, then for $t > 0$,

$$\mathbb{P}(\lambda_{\max}(Z - M) \geq t) \leq q \exp\left(-\frac{t^2}{32b(\nu + t)}\right). \quad (4.3)$$

The same bound holds for $\mathbb{P}(\lambda_{\min}(Z - M) \leq -t)$.

Proof. For a fixed self-adjoint matrix H and $|\theta|b \leq 1/16$, consider

$$\Psi(x) = \text{tr} \exp\left(H + \theta \sum_j x_j A_j + 8\theta^2 \sum_j x_j(1 - x_j) A_j^2\right).$$

We show that Ψ is concave on every allowed line. Write the matrix in the exponential as $Q(t)$ on a line $x + th$. Then

$$Q''(t) = -16\theta^2 \sum_j h_j^2 A_j^2,$$

whereas

$$Q'(t) = \theta \sum_j h_j (A_j + 8\theta(1 - 2x_j - 2th_j) A_j^2).$$

The square of a sum of at most four self-adjoint matrices is at most four times the sum of their squares. Since the coordinates remain in $[0, 1]$,

$$Q'(t)^2 \preceq 4\theta^2(1 + 8|\theta|b)^2 \sum_j h_j^2 A_j^2 \preceq 9\theta^2 \sum_j h_j^2 A_j^2.$$

Thus $Q'' + Q'^2 \preceq 0$.

For a twice differentiable self-adjoint matrix function,

$$\frac{d^2}{dt^2} \text{tr} e^{Q(t)} \leq \text{tr}(e^{Q(t)}(Q''(t) + Q'(t)^2)). \quad (4.4)$$

To verify this, diagonalize $Q(t)$. In the second derivative, the coefficient of $|Q'_{ij}|^2$ is the logarithmic mean $(e^{q_i} - e^{q_j})/(q_i - q_j)$, at most $(e^{q_i} + e^{q_j})/2$. The diagonal coefficients agree. This proves (4.4) and hence concavity.

It follows that $\Psi(x^{(j)})$ is a supermartingale. For the upper tail, take $\theta > 0$ and $H = -\theta M - \theta t I$. Since

$$\sum_j x_j^{(0)}(1 - x_j^{(0)}) A_j^2 \preceq bM \preceq b\nu I,$$

the initial potential is at most $q \exp(-\theta t + 8\theta^2 b\nu)$. The event $\lambda_{\max}(Z - M) \geq t$ forces the terminal potential to be at least one, since its quadratic correction is positive semidefinite. If $t \leq \nu$, use $\theta = t/(16b\nu)$; otherwise use $\theta = 1/(16b)$. These choices imply (4.3). Reversing the sign of the linear term gives the lower tail. $\square$

4.3 Signed fractional circulations

Replace every vertex i of a signed multigraph by states i^+, i^- . An edge ij , with signs s at i and t at j , gives twin arcs

$$i^s \longrightarrow j^{-t}, \quad j^t \longrightarrow i^{-s}. \quad (4.5)$$

Parallel twins and directed loops are retained. Assign the two twins the same lower and upper capacities.

Lemma 4.4. *If this directed graph admits a feasible circulation, then the signed graph admits a fractional vector f satisfying $\mathbf{B}f = 0$ and the corresponding edge capacities.*

Proof. If g is a circulation, define $g^*(a) = g(\bar{a})$, where $\bar{a}$ is the twin of a . Its divergence at i^+ is the negative of the divergence of g at i^- , so g^* is also a circulation. The average $(g + g^*)/2$ is feasible and gives twins equal values. Let f_e be their common value. Its divergence at i^+ is $(\mathbf{B}f)_i$, proving the assertion. $\square$

We use the usual circulation criterion

$$\ell(\delta^-(J)) \leq u(\delta^+(J)) \quad (J \subseteq V(\mathcal{D})). \quad (4.6)$$

For completeness, subtract the lower-capacity vector. Its vertex divergences prescribe demands for the remaining nonnegative flow. Add a source and sink for positive and negative demands, with their prescribed capacities. Max-flow/min-cut gives a flow saturating these arcs exactly when (4.6) holds. Adding back the lower-capacity vector proves sufficiency; necessity follows by summing flow conservation over a shore.

The conclusion of Lemma 4.4 is only fractional. An integral circulation need not agree on twins. We will apply Lemma 4.1 to the physical edges after symmetrization, not project an integral lifted selection.

5 A balanced weighted partition

We return to $X = \text{Cay}(G, S)$. Throughout this section $d/L^{12} \rightarrow \infty$, which is weaker than (1.1). We first obtain a labelled template and then allocate the vertices among a family of its translates.

5.1 A template with well-represented labels

For a labelled subgraph F of order m , put

$$f_s(F) = \frac{1}{m} |\{x \in V(F) : \{x, xs\} \in E(F)\}|.$$

Then $f_s = f_{s^{-1}}$ and $\sum_{s \in S} f_s = d(F)$. An inverse label class $\{s, s^{-1}\}$ has one or two elements, and its physical edges have maximum degree at most two.

Lemma 5.1 (Penalized extraction). *There are disjoint sets $A^+, A^- \subseteq G$ and a symmetric $T \subseteq S$ such that $F = \text{Cay}(G, T)[A^+, A^-]$ is connected and*

$$cd \leq d_F(v) \leq d, \quad 1 - \lambda_2(N_F) \geq cL^{-2}, \quad f_s(F) \geq 1/8 \ (s \in T), \quad |T| \geq cd. \quad (5.1)$$

In particular, $m = |V(F)| \geq cd$.

Proof. Put $\eta = 1/L$. Among all disjoint A^+, A^- and symmetric label sets T , maximize

$$\frac{d(F) - |T|/8}{(|A^+| + |A^-|)^\eta}, \quad F = \text{Cay}(G, T)[A^+, A^-].$$

A cut containing at least half the edges of X , with $T = S$, gives value at least $3d/(8e)$. Write $m = |V(F)|$, $R = d(F)$ and $q = R - |T|/8$. Then $q \geq 3d/(8e)$ and $R \geq cd$.

Deleting a vertex of degree less than $R/2$ increases the average degree and decreases the denominator. Hence $\delta(F) \geq R/2$. If $f_s < 1/8$, deleting its inverse class of size $j \in \{1, 2\}$ decreases the average degree by jf_s and the penalty by $j/8$, increasing the objective. Thus $f_s \geq 1/8$ for every retained label.

For $B \subseteq V(F)$ of size $a \leq m/2$, let $b = m - a$. Comparing with the two restrictions of the bipartite template gives

$$2e_F(B, B^c) \geq q \left(m - \frac{a^{1+\eta} + b^{1+\eta}}{m^\eta} \right).$$

For $x = a/m \leq 1/2$,

$$1 - x^{1+\eta} - (1 - x)^{1+\eta} \geq x(1 - x^\eta) \geq x(1 - 2^{-\eta}) \geq c\eta x.$$

Therefore $e_F(B, B^c) \geq cd|B|/L$. This proves connectedness, and comparable degrees together with Cheeger's inequality give the asserted spectral gap. Finally, $|T| \geq \sum_{s \in T} f_s = R \geq cd$. $\square$

Set $U = \langle T \rangle$ and $u = |U|$. Since F is connected, a translate of it lies inside U ; fix that translate. Every other translate lies in one left U -coset. If X is bipartite, choose the signs of each translated template to agree with the global bipartition. Otherwise, keep its inherited local signs $s_i(v) \in \{+1, -1\}$.

The only connectivity input about vertex-transitive graphs that we use is the following consequence of Watkins's theorem [8]: a connected vertex-transitive graph of degree k has vertex connectivity at least $k/2$. This weaker form suffices for all applications below.

5.2 Fractional allocation

Choose a sufficiently large constant A_0 and put

$$\lambda = A_0 L, \quad \rho = \lambda^{-1}, \quad R = d(F).$$

Independently retain each indexed translate gF , $g \in G$, with probability λ/m . Coincident translated graphs keep their distinct indices. Every vertex lies in exactly m indexed translates. A physical T -edge of inverse class represented by s lies in exactly mf_s indexed translates. For a non-involutory label this follows by orienting the edge by s ; for an involution each physical occurrence has two endpoint identifications.

Chernoff bounds give a family $F_1, \dots, F_t$ for which every vertex lies in at most 2λ copies and

$$\frac{1}{2}\lambda f_s \leq c_e \leq 2\lambda f_s \tag{5.2}$$

for every physical T -edge e of class s , where c_e counts its selected occurrences. The failure probability is at most n^{-30} after choosing A_0 large, because $f_s \geq 1/8$ and there are at most n^2 tests. Fix this cover. In particular, every T -edge is covered.

Give the occurrence of e in F_i weight $w_{i,e} = f_s/c_e$, and define

$$x_{v,i} = \frac{1}{R} \sum_{\substack{e \in E(F_i) \\ v \in e}} w_{i,e}. \tag{5.3}$$

There are two exact identities:

$$\sum_i x_{v,i} = 1 \quad (v \in G), \quad \sum_v s_i(v) x_{v,i} = 0 \quad (i \in [t]). \tag{5.4}$$

For the first, the total weight over all occurrences of one physical s -edge is f_s , and every vertex has one incident edge for every $s \in T$. For the second, every edge occurrence contributes once to each local side. By (5.1) and (5.2),

$$\frac{1}{2}\rho \leq w_{i,e} \leq 2\rho, \quad c\rho \leq x_{v,i} \leq C\rho. \quad (5.5)$$

Only constant-factor bounds are required.

Lemma 5.2. *In the incidence graph between original vertices and copy indices within one U -coset, every cut having copy indices on both shores has at least cd crossing incidences.*

Proof. Partition the copy indices between the shores and let B be the original vertices belonging to copies on both sides. If $|B| < cd$ for a sufficiently small c , every copy has a vertex outside B , since $m \geq cd$. Hence there are vertices belonging exclusively to each family. No T -edge joins these two exclusive sets: a selected copy covering that edge would make at least one endpoint mixed. Removing B would therefore disconnect $\text{Cay}(U, T)$, contrary to its vertex connectivity $\Omega(d)$. Thus $|B| \geq cd$. Every mixed vertex contributes a crossing incidence in any extension of the copy partition to the original vertices. $\square$

The incidence graph is connected within each coset, since it covers all edges of the connected graph $\text{Cay}(U, T)$. Assign signs $+1$ at the original-vertex end of an ownership incidence and $s_i(v)$ at its copy end.

If u is even, use the fractional vector $\pi_{v,i} = x_{v,i}$, with demands one at original vertices and zero at copies. If u is odd, append one auxiliary node to each coset and put

$$\theta = 1 - 1/u, \quad \pi_{v,i} = \theta x_{v,i}, \quad z_v = 1/u. \quad (5.6)$$

Give the reservation incidence $(v, *)$ sign $+1$ at both ends, and demand one at the auxiliary node. The equations are

$$\sum_i \pi_{v,i} + z_v = 1, \quad \sum_v s_i(v) \pi_{v,i} = 0, \quad \sum_{v \in H} z_v = 1 \quad (H \text{ a } U\text{-coset}).$$

In the even case take $\theta = 1$ and omit the reservation variables. Total demand is even in every component: it is u or $u + 1$, respectively.

We check Lemma 4.1. Every coordinate is below $1/2$ for sufficiently large n , so its slack equals its value. A cut splitting the copy nodes has slack $\Omega(\rho d) > 1$, by Lemma 5.2. For any other nontrivial cut choose a shore containing no copy nodes, and let A be its original vertices. Without an auxiliary node in that shore its cut weight is $|A|$. If the auxiliary node is present, the weight is

$$\theta|A| + (u - |A|)/u = 1 + (1 - 2/u)|A| \geq 1.$$

This proves the slack-cut condition, including the even case with no auxiliary node.

Apply signed integrality and then Lemma 4.2, independently in different cosets. The output assigns each original vertex either to exactly one copy or to its reservation node. Every assigned part is exactly balanced. In the odd case there is exactly one reserved vertex w_H in every coset, and mean preservation gives

$$\mathbb{P}(w_H = v) = 1/u \quad (v \in H). \quad (5.7)$$

These reservations are independent across cosets before any conditioning on success of the concentration estimates. Let W be the reserved set, or the empty set when u is even.

5.3 Weights that compensate for the allocation

For every edge uv of the full copy F_i , define in advance

$$a_{i,uv} = \frac{\kappa \rho w_{i,uv}}{\pi_{u,i} \pi_{v,i}}, \quad D = \frac{\kappa \rho R}{\theta}, \quad (5.8)$$

where $\kappa > 0$ is a sufficiently small fixed constant. The comparison (5.5) implies

$$\omega \leq a_{i,uv} \leq 1 \quad (5.9)$$

for some fixed $\omega > 0$. Most importantly, for every vertex of a full copy,

$$\sum_u \pi_{u,i} a_{i,uv} = \frac{\kappa \rho}{\pi_{v,i}} \sum_u w_{i,uv} = \frac{\kappa \rho R}{\theta} = D. \quad (5.10)$$

Thus weighted degrees have exactly the same initial mean, despite the unequal ownership probabilities.

Let $X_{v,i}$ be the final ownership indicators, let $V_i = \{v : X_{v,i} = 1\}$, and give $H_i = F_i[V_i]$ the weights (5.8). Scalar applications of Lemma 4.3 give, simultaneously,

$$|V_i| = \Theta(\rho m), \quad \sum_u X_{u,i} a_{i,uv} = (1 \pm \eta) D, \quad |N_{F_i}(v) \cap V_i| = \Theta(d/\lambda), \quad (5.11)$$

for every full-copy vertex v , whether or not it is assigned to i . Here η can be a sufficiently small fixed multiple of L^{-2} . The degree-tail exponent is $c\eta^2 D$, and the size and support-count tails are at least as strong for the uses above.

We give the spectral calculation, since scalar concentration alone is not sufficient. Let B_i be the normalized adjacency matrix of the full copy with weights $w_{i,e}$. These weights are comparable to ρ , so (2.1) and the template gap give $1 - \lambda_2(B_i) \geq \sigma_0 = cL^{-2}$. Their weighted degrees are $R\pi_{v,i}/\theta$. Therefore the allocated adjacency matrix normalized by the fixed reference degree D is, on retained coordinates,

$$Q_i P_i^{-1/2} B_i P_i^{-1/2} Q_i, \quad P_i = \text{diag}(\pi_{v,i}), \quad (5.12)$$

where Q_i projects onto V_i . This identity follows entry by entry from (5.8).

Write $B_i = u_i u_i^\top + K_i$ and let $K_{i,+}$ be the positive part. By (2.4), $(K_{i,+})_{vv} \leq C d^{-1/2}$. For

$$A_v = \pi_{v,i}^{-1} K_{i,+}^{1/2} e_v e_v^\top K_{i,+}^{1/2},$$

we have

$$\sum_v \pi_{v,i} A_v = K_{i,+}, \quad \|A_v\| \leq C\lambda/\sqrt{d}. \quad (5.13)$$

Apply Lemma 4.3 with deviation $\sigma_0/8$. Since $\|K_{i,+}\| \leq 1 - \sigma_0$, except with probability

$$m \exp(-c\sigma_0^2 \sqrt{d}/\lambda) \quad (5.14)$$

we obtain

$$\|K_{i,+}^{1/2} P_i^{-1/2} Q_i P_i^{-1/2} K_{i,+}^{1/2}\| \leq 1 - 7\sigma_0/8.$$

Its nonzero eigenvalues agree with those of $Q_i P_i^{-1/2} K_{i,+} P_i^{-1/2} Q_i$. Since $K_i \preceq K_{i,+}$, removing the rank-one part from (5.12) and applying interlacing bounds its second eigenvalue by $1 - 7\sigma_0/8$. By (5.11), the reference-normalized matrix has norm $1 + O(\eta)$. Changing reference degrees to the actual weighted degrees perturbs it by $O(\eta)$. Taking η small compared with σ_0 leaves gap at least $\sigma_0/2$.

There are at most n copies and n^2 full-copy vertex tests. Since $\lambda = \Theta(L)$, $D = \Theta(d/L)$ and $d/L^{12} \rightarrow \infty$, both $\sigma_0^2 D \gg L$ and $\sigma_0^2 \sqrt{d}/\lambda \gg L$. The scalar and matrix estimates therefore hold simultaneously with probability $1 - o(1)$.

We have proved the following partition statement.

Proposition 5.3 (Weighted partition). *There are indexed translates F_i and a partition*

$$G \setminus W = V_1 \sqcup \cdots \sqcup V_t, \quad V_i = A_i \sqcup B_i \subseteq V(F_i), \quad |A_i| = |B_i|,$$

such that every vertex lies in at most 2λ full copies and every physical T -edge is covered. The weights (5.8) belong to $[\omega, 1]$, and, with $\sigma = cL^{-2}$ and $D = \Theta(d/L)$,

$$|V_i| = \Theta(m/\lambda), \quad d_{H_i}(v) = (1 \pm c_1\sigma)D, \quad 1 - \lambda_2(N_{H_i}) \geq \sigma. \quad (5.15)$$

The constant c_1 can be chosen sufficiently small for Theorem 3.2. For every full-copy vertex,

$$cd/\lambda \leq |N_{F_i}(v) \cap V_i| \leq Cd/\lambda. \quad (5.16)$$

There is one reserved vertex in each U -coset when u is odd, and none otherwise. In the underlying rounding experiment the reservations have the independent uniform distribution (5.7).

The distributional statement concerns the experiment before conditioning on all the successful events. In the next section we estimate additional failures in that same experiment and use a union bound, rather than assume that the reservations remain uniform after conditioning.

6 A sparse connecting system

The partition is not preserved by translations. The full copies still are, and we use them to prove a cut bound before choosing the connecting edges. In this section assume (1.1), and write $\Lambda = \log L$.

Proposition 6.1 (Connecting system). *For the weighted partition in Proposition 5.3, there is a physical matching M on $G \setminus W$ and a path $P_H = a_H w_H b_H$ for each reserved vertex, such that all their endpoints are distinct, the paths are mutually disjoint, and the endpoints of the paths avoid $V(M)$. Their contraction on the parts is connected and signed-balanced. If E_i is the endpoint set in V_i , then, for any sufficiently small fixed $a > 0$, the choices can be made so that*

$$|E_i| \leq a\Lambda|V_i|/L^7, \quad |N_{F_i}(v) \cap V_i \cap E_i| \leq aD/L^6 \quad (v \in V(F_i)). \quad (6.1)$$

Every E_i is nonempty. A path P_H is represented in the contraction by a loop carrying the local signs of its endpoints.

6.1 Sampling labels and reserving vertices

Independently of the allocation experiment, retain every inverse label class of T with probability t_0/d , where

$$t_0 = A_1\lambda L = \Theta(L^2)$$

and A_1 is a sufficiently large constant. Let F_0 contain the retained edges of F . Conditional on a successful allocation, (5.16) implies

$$|N_{g_i F_0}(v) \cap V_i| \geq ct_0/\lambda \quad (i \in [t], v \in V(F_i)) \quad (6.2)$$

with probability $1 - o(1)$. Indeed, each label-class coin contributes at most two neighbours at a fixed vertex, its mean is $\Omega(t_0/\lambda) = \Omega(L)$, and a weighted Chernoff bound handles all n^2 tests. Also at most $2t_0$ inverse classes are retained with high probability.

Add a fixed symmetric set of $O(L)$ original labels generating G . Such a set is obtained by repeatedly adjoining a label outside the subgroup generated so far; its order at least doubles at each step. In the nonbipartite case choose the added labels so that their lifts $(s, 1)$ generate $G \times C_2$. These lifts generate the whole product: otherwise their subgroup is the graph of a

homomorphism $G \rightarrow C_2$ taking every s to 1, which is precisely a bipartition of X . The same doubling argument then uses $O(L)$ labels.

Let X_0 be the Cayley graph on the union of these added labels and the sampled labels. It is regular of degree

$$k = O(L^2), \quad (6.3)$$

and is connected; its double cover is connected in the nonbipartite case.

We need two facts about cuts of cosets. We state also the cut-counting input, used again for the matching. By the Nash-Williams–Tutte tree-packing theorem, a multigraph of minimum cut at least μ has $\lfloor \mu/2 \rfloor$ edge-disjoint spanning trees; see [4]. It follows that, in a graph with at most $2n$ vertices and $2n^2$ physical edges, the number of cuts of size less than $(j+1)\mu$ is at most

$$(2n)^{C(j+1)}. \quad (6.4)$$

To prove this consequence, one packed tree meets such a cut in at most $3(j+1)$ edges, for μ large. The crossing edges of a tree determine the cut up to complementation. Choose the tree and its crossing set. This gives (6.4); if the allowed crossing size exceeds the tree size, the bound 2^{2n} gives the same conclusion.

Lemma 6.2 (Coset cuts). *In a connected Cayley graph on G , every nontrivial cut whose shore is a union of left U -cosets has at least $u/2$ physical edges.*

Proof. Choose physical representatives of a spanning tree of the coset quotient, and translate them uniformly by G . Every translated tree spans the cosets and hence crosses the proposed cut. A physical undirected edge has a left-translation orbit of size at least $n/2$, since its stabilizer has size at most two. Its probability of belonging to the translated tree is therefore at most

$$\frac{[G : U] - 1}{n/2} \leq 2/u.$$

The expected number of tree edges in the cut is at least one and at most twice its size divided by u . No normality of U is required. $\square$

Suppose u is odd. In the joint allocation and label-sampling experiment, the reservations satisfy, with probability $1 - o(1)$,

$$\text{every coset-union cut retains at least half its edges in } X_0 - W, \quad (6.5)$$

and

$$e_{X_0}(H \setminus W, W) \leq B_* := CkL \quad (H \text{ a } U\text{-coset}). \quad (6.6)$$

Here is the probability calculation. Condition on the sampled labels, so X_0 is fixed. The reservations are independent uniform vertices of their cosets by (5.7), before conditioning on allocation success. For a coset cut of size z , its lost edges are at most a sum of independent variables in $[0, k]$, with mean $2z/u$. The exponential moment at $1/k$ bounds the probability of losing at least $z/2$ edges by $\exp(-cz/k)$. Lemma 6.2 and (6.4) applied to the coset quotient give a simultaneous bound, since $u \geq cd \gg kL$.

For a fixed coset H , the quantity in (6.6) is at most

$$\sum_K |N_{X_0}(w_K) \cap H|.$$

The summands are independent, at most k , and have total mean k , by regularity and $|H| = u$. The same exponential-moment estimate makes the probability of exceeding CkL at most n^{-30} . Union bound over the cosets proves (6.6).

These estimates are compatible with (6.2): the latter is proved conditional on a successful allocation, whereas the reservation estimates are proved conditional on labels before allocation success is imposed. Each failure probability is small in the same joint experiment. A union bound supplies an outcome satisfying all of them. Fix such an outcome and put $Y = X_0 - W$.

6.2 The cut bound for the allocated parts

In the nonbipartite case consider the ordinary bipartite double cover of Y . The two allocated lifts of part i are

$$V_i^\xi = \{(v, \eta) : v \in V_i, \eta = \xi s_i(v)\}, \quad \xi \in \{+1, -1\}.$$

Contract these $2t$ sets, retaining every lifted edge. They are the two states of part i . In the bipartite case we instead contract $V_1, \dots, V_t$ directly in Y .

Lemma 6.3 (Allocation cuts). *In the nonbipartite case the contracted double cover has minimum cut at least*

$$\mu = cd/L. \tag{6.7}$$

If u is odd, the same holds within any single U -coset, using only edges with both original endpoints in that coset. In the bipartite case the ordinary contraction has minimum cut at least μ .

Proof. We give the lifted argument; the ordinary version is identical with the lifts omitted. Let Z be a nontrivial union of allocated states. A remaining lifted vertex is bad if it belongs to a full lifted copy whose allocated part lies across the cut. Let B be the bad set, and let e_i count the disagreeing vertices in each full lifted copy.

Every disagreeing vertex sends at least ct_0/λ edges into that copy's allocated part, by (6.2). Each such edge crosses the cut. A physical lifted edge lies in at most 2λ full lifted copies. Therefore

$$e_{\tilde{Y}}(Z, Z^c) \geq \frac{ct_0}{\lambda^2} \sum_i e_i \geq \frac{ct_0}{\lambda^2} |B|. \tag{6.8}$$

Choose $b_0 = c_0 d/\lambda$, with c_0 so small that b_0 is less than one tenth of every allocated state size. If $|B| \geq b_0$, then (6.8) gives

$$e_{\tilde{Y}}(Z, Z^c) \geq ct_0 d/\lambda^3 = \Omega(d/L).$$

Suppose $|B| < b_0$. No lifted T -edge joins the two exclusive shores outside B , since that edge is covered by a full lifted copy. The components of the full lifted T -graph are cosets of

$$\tilde{U} = \langle (s, 1) : s \in T \rangle \leq G \times C_2.$$

They have degree $|T| \geq cd$ and vertex connectivity $\Omega(d)$. No vertices are reserved if u is even. If u is odd, $\text{Cay}(U, T)$ is nonbipartite, since a connected regular bipartite graph has even order; hence $\tilde{U} = U \times C_2$ and exactly two vertices are reserved in each lifted component. Removing these vertices and B leaves every component nonempty and connected. Thus the cut differs on at most $|B|$ vertices from a nontrivial union of $\tilde{U}$ -cosets. Both shores are nonempty because each contains an allocated state larger than B .

Before reservation, Lemma 6.2 in the connected double cover gives at least $|\tilde{U}|/2 \geq u/2 = \Omega(d)$ edges in this coset cut. When u is odd, its shores are unions of entire sets $H \times C_2$, and (6.5) preserves a fixed fraction of these edges. When returning from the coset cut to Z , only the added $O(L)$ generator labels can decrease the cut size: all sampled T -labels are internal to the $\tilde{U}$ -cosets. The loss is therefore at most $CL|B|$, not $k|B|$. Since $|B| < c_0 d/\lambda$ and $\lambda = A_0 L$, choosing c_0 small leaves $\Omega(d)$ edges.

For the local assertion in an odd coset, the large- B argument uses only internal sampled T -edges. In the small- B case, its sole lifted T -component remains connected. It cannot meet both exclusive shores, while neither shore can lie entirely in B . This is a contradiction. Finally, in the globally bipartite case no vertices are reserved, and the same ordinary argument uses U -cosets and connected X_0 . $\square$

6.3 Directional balance

Give a physical edge of Y the local sign of each endpoint in its allocated part, and form its two-state directed graph as in (4.5). Its underlying undirected graph is exactly the contracted double cover above. For a nontrivial state set J , put

$$s(J) = |\delta_{\mathcal{D}(Y)}(J)|,$$

counting both directions. Then

$$s(J) \geq \mu, \quad |\delta_{\mathcal{D}(Y)}^+(J)|, |\delta_{\mathcal{D}(Y)}^-(J)| \geq s(J)/3. \quad (6.9)$$

Indeed, write $\sigma_i = \mathbf{1}_{\{i^+ \in J\}} - \mathbf{1}_{\{i^- \in J\}}$. The difference of the two directed cut sizes is

$$\sum_i \sigma_i \sum_{v \in V_i} s_i(v) \deg_Y(v). \quad (6.10)$$

Before reservation this vanishes by k -regularity and $|A_i| = |B_i|$. If u is odd, let $q(J)$ count the cosets containing an index with $\sigma_i \neq 0$. Equation (6.6) bounds the absolute value of (6.10) by $q(J)B_*$. In each such coset the state cut is nontrivial, so the local assertion of Lemma 6.3 gives

$$s(J) \geq q(J)\mu. \quad (6.11)$$

Since $B_* = O(L^3) = o(\mu)$, the two directions each contain at least a third of the cut.

For a globally bipartite graph, orient every physical edge from the global positive side to the negative side and contract the parts. The resulting ordinary directed graph is balanced by regularity and exact part balance, including its directed loops. Every nontrivial cut has its two directions equal and total size at least μ . We use this directed graph in place of the two-state graph in that case.

6.4 A physical matching with concentrated cut counts

Lemma 6.4 (Mutual nominations). *Let Y have maximum degree at most k , and let $0 \leq p \leq k^{-2}$. There is a random matching R with $\mathbb{P}(e \in R) = p$ for every edge, such that its edge indicators are negatively associated. Independently marking the edges preserves this property for the marked indicators and, separately, for the unmarked indicators. Weighted counts with coefficients in $[0, 2]$ satisfy ordinary Chernoff bounds.*

Proof. Independently at each vertex, nominate one incident edge: each particular edge is nominated with probability $\sqrt{p}$, and nothing is nominated with the remaining probability. The assumption ensures that these probabilities sum to at most one. Retain an edge when both its endpoints nominate it. The retained edges form a matching and have probability p .

A family is negatively associated if increasing functions of disjoint coordinate sets have nonpositive covariance. The nomination indicators at one vertex have this property. On their one-hot support, each increasing function is its value at zero plus a nonnegative increment, and increments depending on two disjoint coordinate sets cannot both be nonzero. Their covariance is therefore nonpositive.

Independent unions preserve negative association: condition on one family, use the property for the other, then apply it to the two resulting increasing conditional expectations. Increasing functions of disjoint coordinate groups preserve it directly from the definition. Each retained-edge indicator is the product of its two incidence indicators, and the incidence-coordinate groups of distinct edges are disjoint, even when the edges share a vertex. This proves negative association of the retained indicators. Adding an independent Bernoulli coordinate to each group proves the marked assertion; using its complementary coordinate proves the unmarked assertion. These are the usual closure properties of negative association [3].

For nonnegative monotone factors, repeated use of negative association bounds the exponential moment of a sum by the product of the separate moments. This works for both signs of the exponent, since two decreasing functions also have nonpositive covariance. The standard exponential-moment proof gives the upper and lower Chernoff tails, including bounded coefficients. $\square$

Apply the lemma to Y with

$$p = A_2 L / \mu = O(L^2 / d). \quad (6.12)$$

It is valid since $pk^2 = O(L^6 / d) = o(1)$. Let R be the resulting matching, and mark its edges independently with probability $1/128$, giving $Z_0 \subseteq R$.

For any fixed state cut of original size s , a physical edge contributes a weight in $\{0, 1, 2\}$ to its total or directional count. Lemma 6.4 and (6.9) give fixed-cut failure probability at most $C \exp(-cps)$ for the bounds

$$|\delta_{\mathcal{D}(R)}^\pm(J)| \geq ps/12, \quad |\delta_{\mathcal{D}(R)}(J)| \leq 2ps, \quad (6.13)$$

$$ps/512 \leq |\delta_{\mathcal{D}(Z_0)}(J)|, \quad |\delta_{\mathcal{D}(Z_0)}(J)| \leq ps/64. \quad (6.14)$$

The two physical lifts are not sampled independently; they are two contributions of one edge indicator. Cut counting (6.4), with $p\mu = A_2 L$, makes the sum of all failure probabilities $o(1)$ for A_2 sufficiently large. In particular the marked contraction is connected. Applying these estimates to a state cut containing both states of every chosen part gives

$$|\delta_{R \setminus Z_0}(I)| \geq p\mu/16 \quad (6.15)$$

for every nontrivial ordinary part cut I . In the globally bipartite case the same argument applies to its ordinary directed cuts, with the same generous constants.

For any fixed vertex set A , the number of matching endpoints in A is

$$\sum_{e \in E(Y)} |e \cap A| \mathbf{1}_{\{e \in R\}},$$

with mean at most $pk|A|$. Since

$$pk = O(L^4 / d) = O(\Lambda / L^9), \quad (6.16)$$

we can impose simultaneously

$$|V(R) \cap V_i| \leq \frac{a\Lambda|V_i|}{2L^7}, \quad |V(R) \cap N_{F_i}(v) \cap V_i| \leq \frac{aD}{2L^6}. \quad (6.17)$$

The expectations are smaller than these thresholds by factors tending to zero. The least part threshold is $\Omega(L^5)$, since $|V_i| \geq cd/L$; the least neighbourhood threshold is $\Omega(L^6/\Lambda)$. Chernoff tails therefore allow a union bound over all parts and full-copy neighbourhoods. Fix an outcome satisfying all the cut and endpoint estimates.

6.5 Inserting the reserved vertices and choosing an integral connector

For every reserved vertex w_H , choose a full copy F_i containing it. By (5.16), it has at least cd/λ neighbours in V_i . By (6.17), only $o(d/\lambda)$ of those neighbours are endpoints of R . Choose distinct unused neighbours a_H, b_H and form $P_H = a_H w_H b_H$.

The paths for different cosets are disjoint and their endpoints avoid $V(R)$. Both endpoints of a path have the same local sign, opposite to $s_i(w_H)$. Regard it as a formal same-sign loop at part i , and let $\mathcal{P}$ be the family of these loops. There is at most one in each coset and at most one in each part. For any state cut J in the nonbipartite case,

$$|\delta_{\mathcal{D}(\mathcal{P})}(J)| \leq 2q(J) \leq 2s(J)/\mu, \quad (6.18)$$

since a loop crosses only when exactly one state of its part is selected. The arcs of a formal path-loop encode endpoint signs; they are not asserted to be physical lifts of its two-edge path. In the bipartite case $\mathcal{P}$ is empty.

Set $\varepsilon = 1/256$. In the two-state graph of $R \cup \mathcal{P}$, give arcs of $Z_0 \cup \mathcal{P}$ lower and upper capacity one, and give all remaining arcs capacities $[\varepsilon, 1 - \varepsilon]$. For a nontrivial state set J , equations (6.13)–(6.18), with $s = s(J)$, give

$$\begin{aligned} u(\delta^+(J)) - \ell(\delta^-(J)) &\geq (1 - \varepsilon)|\delta_{\mathcal{D}(R)}^+(J)| - \varepsilon|\delta_{\mathcal{D}(R)}^-(J)| - |\delta_{\mathcal{D}(Z_0)}(J)| - |\delta_{\mathcal{D}(\mathcal{P})}(J)| \\ &\geq \left(\frac{1 - \varepsilon}{12} - 2\varepsilon - \frac{1}{64}\right)ps - \frac{2s}{\mu} \geq \frac{ps}{20} - \frac{2s}{\mu} > 0. \end{aligned}$$

The last inequality uses $p\mu = A_2L \gg 1$. Empty and full cuts satisfy the criterion trivially. Thus (4.6) gives a feasible ordinary circulation. Lemma 4.4 symmetrizes it to a fractional signed vector on the physical edges and formal loops, satisfying

$$\mathbf{B}f = 0, \quad f_e = 1 \ (e \in Z_0 \cup \mathcal{P}), \quad \varepsilon \leq f_e \leq 1 - \varepsilon \ (e \in R \setminus Z_0). \quad (6.19)$$

In the globally bipartite case use the ordinary directed graph, with no loops from $\mathcal{P}$; the same cut inequalities give (6.19) directly.

For any nontrivial ordinary part cut,

$$\sum_{e \in \delta(I)} \min\{f_e, 1 - f_e\} \geq \varepsilon|\delta_{R \setminus Z_0}(I)| \geq \varepsilon p\mu/16 > 1.$$

The signed demand is zero, so every component has even total demand. Lemma 4.1 puts f in the convex hull of binary balanced selections. Every binary selection with positive coefficient contains $Z_0 \cup \mathcal{P}$, since those coordinates of f equal one. Choose one, and denote its selected physical edges by $M \subseteq R$.

The contraction of $M \cup \mathcal{P}$ is connected because it contains the marked contraction, and signed-balanced by construction. The physical edges remain a matching. Each part receives at most two extra endpoints from $\mathcal{P}$, so (6.17) implies (6.1). Every part is incident with the connecting system when there are at least two parts. This is the case here: (5.4) and $x_{v,i} \leq C/\lambda$ force at least $c\lambda$ selected copies, and all their allocated parts are nonempty. This proves Proposition 6.1.

7 Completion of the proposed proof

Proof of Theorem 1.1. We now check the quantitative conditions together. Let $L = \log n$, $\Lambda = \log L$, and suppose

$$d \geq C_0 L^{13}/\Lambda,$$

where C_0 is sufficiently large. Obtain the template from Lemma 5.1, the weighted partition from Proposition 5.3, and the connecting system from Proposition 6.1.

The principal scales are

$$\begin{aligned} \lambda &= \Theta(L), & D &= \Theta(d/L), & \sigma &= \Theta(L^{-2}), \\ g &= \Theta(L/\Lambda), & k &= O(L^2), & \mu &= \Omega(d/L), \\ p &= O(L^2/d), & pk &= O(L^4/d), & w/N &= \Theta(\Lambda/L^7). \end{aligned} \quad (7.1)$$

We explain each restriction rather than appeal to an unspecified separation of scales.

First, the allocation requires $d/L^{12} \rightarrow \infty$, which follows because $L/\Lambda \rightarrow \infty$. More explicitly, the scalar degree exponent is at least a constant times

$$\sigma^2 D \geq cC_0 L^8/\Lambda \gg L,$$

and its matrix exponent is at least

$$\frac{\sigma^2 \sqrt{d}}{\lambda} \geq c \sqrt{C_0} \frac{L^{3/2}}{\sqrt{\Lambda}} \gg L.$$

The constant-factor translate cover and direct-neighbour label sampling each have exponents that are sufficiently large fixed multiples of L . The reservation estimates use $u \geq cd \gg kL$, and their boundary error is $B_* = O(L^3) = o(\mu)$.

The mutual-nomination matching is valid because $pk^2 = O(L^6/d) = o(1)$. Its all-cuts bound uses $p\mu = A_2 L$, with A_2 chosen after the cut-count constant. The endpoint density satisfies

$$pk = O(\Lambda/L^9) = o(\Lambda/L^7), \quad pk = o(L^{-6}),$$

which justifies the two different endpoint bounds in (6.1). Their thresholds are at least cL^5 and cL^6/Λ , respectively, so all required neighbourhood and part counts concentrate simultaneously.

For each part, delete from H_i any physical edge selected in M with both endpoints in that part. Call the remaining weighted graph H'_i . Since M is a matching and weights are at most one, every degree decreases by at most one. The deleted adjacency has norm $O(1/D)$ in the old normalization. Lemma 2.1 therefore preserves (5.15), after decreasing the gap constant and increasing the relative error by $O(1/D)$. Choose the original error sufficiently small, so H'_i meets the near-regularity requirement of Theorem 3.2. Its positive weights still lie in $[\omega, 1]$.

Write $N_i = |V_i|/2$, and apply that theorem with $L = \log n$ and $\sigma = c_* L^{-2}$ for a sufficiently small fixed c_* . Take $g = C_g L/\Lambda$, with C_g sufficiently large. Since $D \geq cC_0 L^{12}/\Lambda$,

$$\sigma^{5/2} D/L \geq cC_0 L^6/\Lambda.$$

Its logarithm is at least 5Λ for sufficiently large n . Thus the cycle-length condition in (3.1) follows by choosing C_g large. The other local degree condition is

$$D \geq C\sigma^{-9/2} L^2 g = O(L^{12}/\Lambda),$$

which follows by choosing C_0 after C_g and the fixed template constants.

Signed balance says that E_i has equally many vertices on the two local sides. Put $\ell_i = |E_i|/2$. Proposition 6.1 gives

$$\ell_i \leq a\Lambda N_i/L^7, \quad \max_v d_{H'_i}(v, E_i) \leq aD/L^6.$$

The latter uses that each positive edge weight is at most one. Since

$$\frac{\sigma^{5/2}}{Lg} = \Theta(\Lambda/L^7), \quad \frac{\sigma^{5/2}}{L} = \Theta(L^{-6}),$$

choosing a sufficiently small verifies (3.2). Every endpoint set is nonempty. Therefore $(\text{supp } H'_i, E_i)$ is matching-absorbing for every i .

The matching M and the reserved-vertex paths have mutually distinct endpoints and disjoint interiors partitioning W . Their contraction is connected, and no edge used internally in H'_i belongs to the connecting matching. Lemma 3.8 now gives a Hamilton cycle in the support of the original graph X . This completes the proposed proof of Theorem 1.1. $\square$

The logarithmic exponent

The final degree condition comes from the local matching layers, not the global connector. The router width is of order

$$w/N = \frac{\sigma^{5/2}}{Lg},$$

while two-stage sampling requires matching-layer degree at least $C\sigma^{-2}L$. Thus

$$D \geq C\sigma^{-9/2}L^2g.$$

At $\sigma = \Theta(L^{-2})$ and $g = \Theta(L/\log L)$, this is $D \geq CL^{12}/\log L$. The logarithmic overlap gives $D = \Theta(d/L)$ and hence the exponent in (1.1). Equivalently, the three factors are

$$\underbrace{L}_{\text{allocation loss}} \cdot \underbrace{L^7/\log L}_{\text{inverse router width}} \cdot \underbrace{L^5}_{\text{matching-layer degree}} = \frac{L^{13}}{\log L}.$$

This explains the exponent in the present argument; it is not an optimality assertion for Cayley Hamiltonicity.

Scope and sources

The manuscript is self-contained with respect to the application-specific arguments: the spectral deletion and connection lemmas, the local absorption construction, signed integrality and swap rounding, the trace-exponential concentration estimate, weighted allocation, cut estimates, and final physical-edge selection have all been included. The classical inputs stated without proof are the normalized Cheeger inequality, matrix Bernstein, Hall's and Haxell's matching theorems, the capacitated matching-polytope description, Watkins's connectivity bound, the tree-packing theorem, and max-flow/min-cut.

The matrix-sampling and router framework is taken from the supplied manuscript of Müyesser [6]; the weighted and matching-absorption adaptations are part of the present candidate argument. The signed allocation machinery originates in the earlier drafts, including [9], and is proved here rather than assumed. The proposed theorem, particularly its local absorption and dependent matrix-rounding steps, still requires independent mathematical review. Consolidating the proofs does not constitute that review.

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